Epigenetics activity key

Methylation can occur

- directly on the DNA (usually cytosine is methylated)
    - blocks transcription if occurs at a promoter (typical pattern)
    - This paper includes an example of intron methylation that increases transcription of the gene – *this is not promoter methylation which always blocks or downregulates transcription.*
- on the histone tails
    - Can "loosen" or "tighten" the DNA

Acetylation can only occur on the histone tails & always "loosens" chromatin (makes it easier for the transcriptional proteins to access the genes

Label	Epigenetic System	Core Epigenetic Focus	Paper Sections to Read
1	**Glucocorticoid Receptor (GR / NR3C1)**	DNA methylation at promoters & CpG shores	**Section 3.1** (DNA methylation changes at the GR in hippocampus and PVN)
2	Arginine Vasopressin (Avp)	Activity-dependent DNA demethylation via MeCP2	**Section 3.2** (DNA methylation changes at Avp in the hypothalamus) **and** relevant Avp discussion in **Section 4**
3	Corticotropin-Releasing Factor (Crf)	Context-dependent promoter methylation	**Section 3.2** (Crf portion only) **and** Crf-related mechanisms in **Section 5.2**
4	FKBP5	Intron demethylation at glucocorticoid response elements	**Section 3.3** (DNA methylation changes at Fkbp5) **and** FKBP5 discussion in **Section 5.2**
5	BDNF	DNA methylation + histone H3K27 methylation	**Section 4** (BDNF paragraphs only) **and** BDNF-related sensory development in **Section 6**
6	Genome-wide glucocorticoid effects	Global DNA methylation & chromatin modifiers	**Section 5.1** (Genome-wide effects of glucocorticoids)

No group is assigned **Section 1 or 2** explicitly
→ those sections function as **background**, not content ownership.

Detailed section key

🔷 Section 3.1 - #1

DNA methylation changes at the GR (NR3C1) in hippocampus and PVN

What is being modified?

- **DNA methylation at specific CpG sites**
- Located in:
    - **Promoter region** of GR (hippocampus)
    - **CpG island shore regions** near promoter (PVN)

What is happening at the DNA level?

the promoter (or promoter-adjacent region) becomes methylated.

Mechanistically:

1. CpG sites in the GR regulatory region gain methyl groups
2. This:
    - **Prevents binding of transcription factors** (e.g., NGFI-A)
    - **Recruits methyl-binding proteins** (e.g., MeCP2)
3. These proteins recruit:
    - HDACs
    - Repressive histone modifiers
4. Local chromatin becomes **less accessible**

What environmental input is affecting the system?

- **Early life stress**, especially:
    - Low maternal care
    - Maternal separation
    - Early adverse caregiving environments
- These inputs alter **stress hormone exposure** during development.

(Explicitly discussed in Section 3.1 in the context of rodent and human studies.)

Where in the genome does this modification occur?

- **Hippocampus:**
    - **Promoter region** of the GR gene
    - Exon 17 (rodents) / Exon 1F (humans)
- **Hypothalamic PVN:**
    - **CpG island shore regions** adjacent to the GR promoter

These are **regulatory regions controlling transcription**, not coding regions.

What does this do to the body long term?

- Reduced GR expression → weaker negative feedback on the HPA axis
- Results in:
  - Prolonged glucocorticoid release
  - Heightened stress reactivity
  - Increased vulnerability to anxiety- and stress-related disorders

These effects **persist into adulthood**, even after the original stressor is gone.

Net effect

- DNA is still wrapped
- **Transcription factors cannot physically access the promoter**
- GR transcription ↓

What the model should show

- Push pins **on DNA near/in the promoter**
- Transcription factor unable to bind
- Possibly a "repressor recruited" label

🔷 **Section 3.2 (Avp portion) #2**

DNA methylation changes at Avp enhancer in PVN

What is being modified?

- **DNA demethylation** at CpG sites
- Located at an **enhancer**, not the promoter

What is happening at the DNA level?

- Early life stress → neural activity
- Neural activity → **phosphorylation of MeCP2**
- Phosphorylated MeCP2:
  - Releases from DNA
  - Stops recruiting DNMTs
- CpGs at enhancer become **demethylated**

What environmental input is affecting the system?

- **Early life stress that increases neural activity** in hypothalamic stress circuits
- This includes:
  - Maternal separation
  - Repeated early stressors

The key input here is **activity**, not just hormone levels.

Where in the genome does this modification occur?

- **Enhancer region** of the *Avp* gene
- Located in **parvocellular neurons of the PVN**
- CpG sites within the enhancer undergo **demethylation**

This is **not promoter methylation**.

What does this do to the body long term?

- Increased Avp expression
- Leads to:
    - Stronger activation of the HPA axis
    - Higher basal and stress-induced corticosterone levels
    - Long-term elevation of stress responsivity

This contributes to a **hyper-reactive stress phenotype**.

Key point

❗ This is **loss of repression**, not active activation.

Net effect

- Enhancer becomes accessible
- Enhancer can now interact with promoter
- Avp transcription ↑

What the model should show

- Push pins **removed from enhancer region**
- Increased accessibility
- Label that enhancer–promoter interaction is now possible

🔷 Section 3.2 (Crf portion) #3

Context-dependent DNA methylation at Crf

What is being modified?

- **DNA methylation at promoter CpGs**
- Brain-region specific (PVN vs hippocampus)

What is happening at the DNA level?

- Stress exposure leads to:
  - Gain *or* loss of promoter methylation depending on context
- Methylation status determines:
  - Whether transcription factors can bind

What environmental input is affecting the system?

- **Type, timing, and context of early life stress**
- Includes:
  - Prenatal stress
  - Postnatal stress
  - Different caregiving environments

The paper emphasizes **stressor-specific effects**.

Where in the genome does this modification occur?

- **Promoter region** of the *Crf* gene
- CpG methylation patterns differ by:
  - Brain region (PVN vs hippocampus)
  - Type of stressor

What does this do to the body long term?

- Fine-tunes Crf expression rather than simply increasing or decreasing it
- Alters:
  - Baseline stress hormone regulation
  - Responsiveness to future stressors

This contributes to **individual differences in stress sensitivity and coping**, not a single uniform outcome.

Important nuance

⚠ This section emphasizes **heterogeneity**:

- Sometimes methylation changes explain expression
- Sometimes expression changes occur without clear methylation changes

Net effect

- Promoter accessibility is **context-dependent**
- Crf expression is **fine-tuned**, not simply on/off

What the model should show

- Promoter with **conditional accessibility**
- TA should accept:
    - Either methylation added or removed
    - As long as students justify it with context

🔷 **Section 3.3 #4**

FKBP5 intronic demethylation

What is being modified?

- **DNA demethylation at intron 7**
- Specifically at **glucocorticoid response elements (GREs)**

What is happening at the DNA level?

1. Glucocorticoids bind GR
2. GR binds GREs in FKBP5 intron
3. This binding:
    - Triggers **local DNA demethylation**
4. Demethylated intron:
    - Becomes more accessible to GR
    - Creates a positive feedback loop

What environmental input is affecting the system?

- **Early life stress combined with glucocorticoid exposure**
- Stress hormones activate GR, which directly interacts with the FKBP5 gene

Where in the genome does this modification occur?

- **Intron 7** of the *FKBP5* gene
- Specifically at **glucocorticoid response elements (GREs)**
- DNA at these sites becomes **demethylated**

This is a key example of **intronic regulatory DNA**.

What does this do to the body long term?

- Increased FKBP5 expression
- FKBP5 inhibits GR function
- Results in:
    - Reduced glucocorticoid receptor sensitivity
    - Prolonged cortisol responses to stress

This creates a **self-reinforcing stress dysregulation loop**.

Key TA clarification

❗ This is **not promoter methylation**
❗ Intronic regulatory DNA is being modified

Net effect

- FKBP5 transcription ↑
- FKBP5 inhibits GR function
- Stress signaling becomes prolonged

What the model should show

- Push pins **removed from intronic region**
- GR TF binding enhanced
- Feedback loop annotation

🔷 Section 4 (BDNF) #5

DNA + histone methylation at Bdnf promoters

What is being modified?

- **DNA methylation** at Bdnf promoters
- **Repressive histone methylation** (H3K27me3)

What is happening at the DNA level?

- Promoter CpGs gain methylation
- Polycomb complexes are recruited
- Histone tails gain repressive methylation
- Chromatin becomes **stably inaccessible**

What environmental input is affecting the system?

- **Early life stress and altered sensory experience**

- Includes:
    - Stress combined with sensory deprivation or threat
    - Altered neural activity during development

Where in the genome does this modification occur?

- **Promoter regions** of the *Bdnf* gene (especially exon IV)
- Associated with:
    - DNA methylation
    - Repressive histone methylation (H3K27me3)

What does this do to the body long term?

- Reduced Bdnf expression
- Leads to:
    - Decreased synaptic plasticity
    - Altered critical period timing
    - Long-term changes in learning, memory, and emotional regulation

These effects shape **how neural circuits develop and function**.

Net effect

- Transcription factors blocked
- Bdnf expression ↓
- Plasticity reduced

What the model should show

- Push pins on promoter
- Binder clilps on histone tails
- Persistent “closed” state

🔷 Section 5.1 (Genome-wide glucocorticoid effects) #6

What is being modified?

- **No single locus**
- Changes in:
    - DNMT expression
    - HDAC/HAT activity

What is happening at the DNA level?

- Hormones change the **probability** that DNA will become accessible or inaccessible elsewhere
- This is **priming**, not direct regulation

What environmental input is affecting the system?

- **Chronic or repeated early glucocorticoid exposure**
- Often driven by early life stress

Where in the genome does this modification occur?

- **No single locus**
- Affects:
    - DNMT expression
    - HDAC/HAT activity
    - Global chromatin accessibility

What does this do to the body long term?

- Alters the **epigenetic landscape broadly**
- Changes how likely genes are to become accessible or repressed later
- Primes the organism for:
    - Heightened stress sensitivity
    - Altered responses across multiple systems

What the model should show

- Multiple regions affected at once
- Increased or decreased likelihood of repression globally

Summary Table (TA quick view)

Section	DNA-level change	TF access?
3.1 GR	Promoter/CpG shore methylation	Blocked
3.2 Avp	Enhancer demethylation	Increased
3.2 Crf	Context-dependent promoter methylation	Variable
3.3 FKBP5	Intron demethylation at GREs	Increased
4 BDNF	Promoter methylation + H3K27me3	Blocked

Section	DNA-level change	TF access?
5.1 Genome-wide	Modifier enzyme levels altered	Globally shifted

One sentence that fixes 90% of confusion

"Ask yourself: *Is this modification blocking proteins from binding DNA here, or allowing them to bind more easily?* That's what the model needs to show."

Student activity 'what to look for's'

Background and Context:

Why would changes in hormone levels or neural activity alone be insufficient to explain effects of early life stress that last for decades? These changes are transient, for their effects to last long term they, or something else must be producing an effect that is not transient.

Why might epigenetic changes in one neural system be sufficient to alter behavior, even if other systems are not directly modified? The various ‘neuronal systems’ (parts of the nervous system) cascade into each other – they are all interconnected and affect each other.

As you read about specific genes and epigenetic marks, how will you decide whether a change represents short-term regulation or long-term developmental programming? 1. When did the change occur? A central tenet of the paper is that long-term changes occur early in development. 2. Is there evidence this change has long-term effects?

Activity (Poster):

If students are actively looking words up and writing them on their poster they are lost. I have to look up lots of words when I read a scientific paper and I have a stupid number of degrees.

Tell students they need to DRAW their affected system, diagram relationships, etc. It does them no good to just write words.

If you see phrases that seem like they are directly from the paper ask students ‘what does that mean?’ Work through it with them in needed then have them rewrite in their own words.

Note: e.g. means ‘for example’ (exempli gratia) these are not exhaustive lists of possibilities.

Activity (Build the Model):

Make sure the specific region that is affected is represented (promoter, intron 7, etc.)

DNA is always wrapped around histones, it is never unwrapped.

“Loose” and “tight” chromatin refers entirely to the accessibility (can the transcription proteins get to the gene)